# HPY413Assignment: 7

# Network traffic monitoring using the Snort intrusion detection system

Chaintouti Maria – 2020030129

Cheiladakis Nikolaos – 2020030140

Part 6:

**1. Describe how Snort uses rules to detect malicious activity:**

Snort uses signature-based detection by comparing traffic against predefined rules, that are consisted of headers (protocol, source/destination) and options (payload/content analysis). Each rule defines conditions such as protocol, IP/port range, and payload patterns. When traffic matches a rule, Snort generates an alert.

**2. Mention at least 6 limitations of signature-based intrusion detection systems like Snort with a small description for each one:**

Limited Effectiveness Against Unknown Threats: Signature-based IDS relies on a predefined database of signatures. Consequently, it struggles to detect novel or previously unseen threats for which no signature exists.

Vulnerability to Signature Manipulation: Malicious actors can circumvent signature-based IDS by obfuscating their attacks or modifying known signatures slightly to evade detection.

False Positives and Negatives**:** Signature-based IDS may generate false positives, where benign activities are erroneously flagged as malicious. Conversely, it can also produce false negatives, failing to detect actual intrusions due to incomplete or outdated signatures.

Inability to Detect Encrypted Traffic**:** As encryption becomes ubiquitous in network communications, signature-based IDS faces challenges in inspecting encrypted traffic effectively.

Maintenance Overhead**:** Maintaining an up-to-date database of signatures requires continuous effort and resources. Moreover, the proliferation of new threats necessitates frequent updates to the signature library.

Performance Overhead: As the number of rules increases, the system's performance may degrade, slowing down traffic analysis and increasing latency.

**3. Pros and Cons of Using Snort in Real-World Scenarios:**

Snort IDS offers several advantages that make it a popular choice for organizations seeking robust network security. Its open-source nature makes it highly cost-effective compared to proprietary solutions, while still providing powerful features like protocol analysis, content searching, and real-time alerting. The system's flexibility and extensibility allow security teams to customize rules and signatures to suit their specific network environments, improving detection accuracy and reducing false positives. Additionally, Snort integrates with various operating systems and benefits from a community-driven rule set that is constantly updated with the latest threat intelligence, keeping it adaptable to evolving threats.

However, Snort does come with its challenges. It requires continuous monitoring and fine-tuning of its rules to remain effective, which can be time-consuming and resource-intensive. For high-traffic networks, performance may degrade due to the processing demands of large rule sets. Additionally, while Snort is scalable, managing it in larger or more complex environments can be challenging without adequate expertise. Its reliance on signature-based detection also means it has limited ability to detect zero-day threats, making it necessary to complement it with other security measures.

Sources for 2, 3:

What Are the Limitations of Signature-Based Intrusion Detection? - Sentinel Over Watch

Getting Started with Snort IDS: Examining The Pros and Cons for Maximizing Network Security | EC-Council Whitepaper